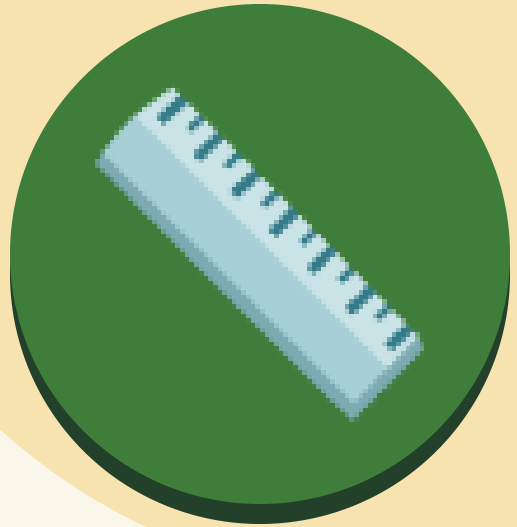
Small Engines Overhaul · Unit 1 · Level 2

Measurement Stations



TODAY'S PLAN

- 👥 Jobs
- ⚠️ Safety
- 📏 Steps
- ✅ Check
- 📝 Reflect

Get ready

Jobs



Mechanic

Does the main engine job with the tools



Tool helper

Gets tools from the board and returns them to staff



Recorder

Takes photos and fills in the chart



Clean-up captain

Wipes the bench, sweeps, checks tools are back

Safety

Be Safe



- 🥽 **Safety glasses** on the whole time.
- 👖 **Coveralls** and closed-toe shoes; tie back long hair.
- 🔧 Use tools **with adult supervision**. Right tool for the job.
- 💔 Measuring tools are **delicate**: no dropping, no forcing.
- 🧼 **Clean up** and wash hands when you finish.

Steps

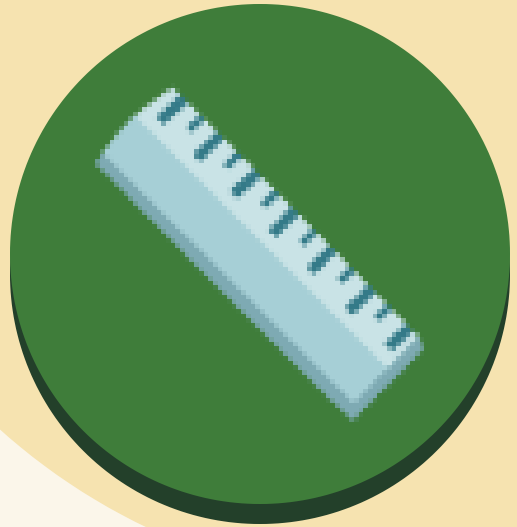
Steps 1 of 2



1. 🧽 Wipe each station part clean.
2. 🗜️ Station 1: measure the rod diameter and length with the caliper.
3. 🔩 Station 2: measure the crankshaft journals with the micrometer.
4. 📄 Station 3: find the feeler blade that fits the sample gap.

Steps

Steps 2 of 2



1. 📏 Station 4: measure a part with a ruler to the nearest 1/16.
2. 🎯 Write measured and spec side by side.
3. 📦 Return tools to their cases.

🃏 Check 1 of 3



Bar 0.500 + dial 0.023 =

Hold up 1, 2, or 3:

1️⃣ 0.523

2️⃣ 0.023

3️⃣ 5.23

✅ Answer · Card 1 of 3



0.523

Add.

🃏 Check 2 of 3



Micrometer sleeve 0.750 + thimble 0.009 =

Hold up 1, 2, or 3:

1️⃣ 0.750

2️⃣ 0.7509

3️⃣ 0.759

✅ Answer · Card 2 of 3



0.759

Add.

🃏 Check 3 of 3



The right feeler blade fits with…

Hold up 1, 2, or 3:

1️⃣ Hammering

2️⃣ Light drag

3️⃣ No contact

✅ Answer · Card 3 of 3



Light drag

Light drag.

Wrap-Up

Clean Up + Reflect



Return and count tools.

Sweep and wipe your bench.

Wash hands, then do the reflection.